MediGuard 2.0 - A smart, secure medicine box with App monitoring

A Project Phase-1 Report

Submitted to the APJ Abdul Kalam Technological University

in partial fulfillment of requirements for the award of degree

Bachelor of Technology

in

Electronics and Communication Engineering

by

ARJUN RAJ A (SPT22EC009)

MUHAMMED SAHAL N (SPT22EC017)

ANAMIKA K (SPT22EC006)

NAVYA AP (SPT22EC018)



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

SREEPATHY INSTITUTE OF MANAGEMENT AND TECHNOLOGY,VAVANOOR

KERALA

October 2025

DEPT. OF ELECTRONICS & COMMUNICATION ENGINEERING

SREEPATHY INSTITUTE OF MANAGEMENT AND TECHNOLOGY

VAVANOOR

2025-26

CERTIFICATE

This is to certify that the report entitled **MediGuard 2.0 - A smart, secure medicine box with App monitoring** submitted by **ARJUN RAJ A** (SPT22EC009) to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Electronics and Communication Engineering is a bonafide record of the main project phase 1 work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Ms. ANJALI ANANTHAN
(Project Guide)
Associate Professor
Dept.of ECE
SIMAT
Vavanoor

Ms. Sreebala P
(Project Coordinator)
Associate Professor
Dept.of ECE
SIMAT
Vavanoor

Dr. VINOD C
Professor and Head
Dept.of ECE
SIMAT
Vavanoor

DECLARATION

We hereby declare that the project report **MediGuard 2.0 - A smart, secure medicine box with App monitoring**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Ms. Anjali Ananthan.

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

ARJUN RAJ A
MUHAMMED SAHAL N
ANAMIKA K
NAVYA AP

Vavanoor
11-10-2025

Abstract

Medication management is a critical aspect of elderly and special-needs patient care, as missed or incorrect doses can result in serious health risks. MediGuard 2.0 is an intelligent medication management system designed to provide secure, accurate, and timely access to prescribed medicines. The system is powered by an ESP32 DevKitC-32UE microcontroller, which controls multiple integrated modules, including a stepper motor-driven rotating tray with six compartments for precise scheduling, an OLED display for real-time guidance, and a DFPlayer Mini with speaker for voice-based alerts. To ensure secure access, the device employs an R307 fingerprint scanner for patients and an RC522 RFID reader for caregivers. An infrared sensor verifies medicine pickup, while a mobile application allows scheduling, language settings, caregiver notifications, and real-time monitoring. The system is supported by a rechargeable lithium-ion battery pack with a BMS and 12V charging, ensuring continuous operation even during power outages. By combining biometric security, automated dispensing, real-time alerts, and remote caregiver connectivity, MediGuard 2.0 significantly enhances medication adherence, improves patient safety, and provides a reliable solution for home healthcare and assisted living environments.

Acknowledgement

I am profoundly grateful to our esteemed Principal, **Dr. S.P. Subramanian**, for granting me this invaluable opportunity to carry out my Main Project – Phase 1. His constant encouragement and support have been truly inspiring.

My sincere thanks go to **Dr. Vinod C**, Professor and Head, Department of Electronics and Communication Engineering, for his valuable guidance, insightful suggestions, and continuous support throughout this phase of the project.

I would like to express my deep appreciation to **Prof. Sushma M**, Department of ECE, for her valuable assistance and coordination during the course of this work. My heartfelt gratitude also goes to **Ms. Anjali Ananthan**, Associate Professor, Department of ECE, for her expert advice, motivation, and consistent support, which greatly contributed to the successful completion of this phase.

I thank the Almighty for His blessings throughout this endeavor. Finally, I extend my sincere appreciation to my friends and family for their encouragement, support, and good wishes during the completion of this work.

ARJUN RAJ A

MUHAMMED SAHAL N

ANAMIKA K

NAVYA AP

Contents

Abstract **i**

Acknowledgement **ii**

List of Figures **v**

1 Introduction **1**

2 Literature Review **2**

3 MEDIGUARD 2.0 - A SMART, SECURE MEDICINE BIX WITH APP MONITORING **20**

3.1 Motivation 20

3.2 Objectives 21

3.3 Methodology 22

3.4 Block diagram 23

3.5 Circuit Diagram 24

3.6 Components used 26

3.6.1 ESP32 DevKitC 32UE 26

3.6.2 FINGERPRINT READER (R307) 27

3.6.3 IR SENSOR (LM393) 27

3.6.4 RFID MODULE (RC522) 28

3.6.5 STEPPER MOTOR (CREALITY 42-40) 29

3.6.6 STEPPER MOTOR DRIVER MODULE (DRV8825) 29

3.6.7 Mini DF PLAYER 30

3.6.8 MINI 5V DIGITAL AMPLIFIER (PAM8403) 31

3.6.9 0.5W SPEAKER (2.5”) 31
3.6.10 OLED DISPLAY MODULE 32
3.6.11 LI-ION BATTERY 33
3.6.12 3S LI-ION BMS 33
3.6.13 12V 1A DC ADAPTER 34
3.6.14 BUCK CONVERTER (LM2596) 35
3.6.15 RTC MODULE (DS3231) 35
3.7 Software used 36

4 Conclusion **37**

References **38**

List of Figures

3.1 Block diagram of MediGuard 2.0 System 23
3.2 Circuit Diagram of Mediguard 2.0 24
3.3 ESP32 DevKitC 32UE 26
3.4 FINGERPRINT READER (R307) 27
3.5 IR SENSOR (LM393) 28
3.6 RFID MODULE (RC522) 28
3.7 STEPPER MOTOR (CREALITY 42-40) 29
3.8 STEPPER MOTOR DRIVER MODULE (DRV8825) 30
3.9 Mini DF PLAYER 30
3.10 MINI 5V DIGITAL AMPLIFIER (PAM8403) 31
3.11 0.5W SPEAKER (2.5”) 32
3.12 OLED DISPLAY MODULE 32
3.13 LI-ION BATTERY 33
3.14 3S LI-ION BMS 34
3.15 12V 1A DC ADAPTER 34
3.16 BUCK CONVERTER (LM2596) 35
3.17 RTC MODULE (DS3231) 35

Chapter 1

Introduction

MediGuard 2.0 is an advanced smart medication management system developed to ensure secure, accurate, and timely access to prescribed medicines, particularly for elderly patients and individuals with special healthcare needs. At its core, the system is powered by an ESP32 DevKitC-32UE microcontroller, which efficiently coordinates all modules and also provides wireless connectivity for app-based monitoring. The device features a stepper motor-driven rotating tray divided into eight compartments, allowing precise alignment of the correct medicine compartment according to scheduled dosages. Scheduling is managed by a DS3231 real-time clock, which can also synchronize with internet time for improved accuracy. For security and controlled access, the system integrates dual authentication: an R307 fingerprint scanner for patients and an RC522 RFID reader for caregivers. An infrared sensor is used to confirm medicine pickup, while an OLED display delivers instructions and real-time information, and a DFPlayer Mini with speaker provides voice alerts and reminders at medication times. The system is powered by a rechargeable 3-cell lithium-ion battery pack, protected by a BMS and charged via 12V DC Adaptor, ensuring uninterrupted use during power outages. A companion mobile application allows caregivers to configure medication schedules, set language preferences, send alerts, and track adherence remotely. With its combination of automation, authentication, voice guidance, and app connectivity, MediGuard 2.0 provides a reliable, user-friendly, and secure solution to enhance patient compliance and reduce health risks associated with missed or incorrect doses.

Chapter 2

Literature Review

D. Satanure, S. Kamlaskar, Y. Vaidya and S. Metkar, "Design and Implementation of a Smart Medicine Reminder Kit for Elderly Care," 2025 Global Conference in Emerging Technology (GINOTECH), PUNE, India, 2025 [1]

Health monitoring and proper medication intake are essential aspects of elderly care, as missing doses or consuming incorrect medication can lead to serious health complications. To address this concern, a smart medicine reminder kit has been developed using an ESP32 microcontroller integrated with a GSM (SIM900A) module. The system is designed to provide timely reminders through LED and buzzer alerts, ensuring that patients are notified at the correct time to take their medication. In addition, the integration of GSM technology allows real-time communication, enabling updates and alerts to be sent directly to caregivers through a mobile application. This ensures that any missed doses or irregularities in medication intake can be monitored remotely, allowing immediate intervention if required. The proposed system is simple yet highly effective in promoting medication adherence and reducing dependency on others. By highlighting the role of ESP32 in wireless connectivity and remote caregiver monitoring, this work provides a foundation for enhancing smart medicine dispensing systems. The concept is highly relevant to MediGuard 2.0, as it aligns with the project's objectives of real-time monitoring, timely alerts, and ensuring patient safety through caregiver–patient interaction and remote supervision.

K. Ramani, P. V. Kumar, N. Jahnavi, A. Karthik, N. Yadav and U. S. Rahul, "Smart Medication Monitoring In Medical IoT," 2023 2nd International Conference on Vision Towards Emerging Trends in Communication and Networking Technologies (ViTECoN), Vellore,India,2023 [2]

The growing integration of Internet of Things (IoT) in healthcare has led to the development of systems that ensure safe and reliable medication intake, particularly for elderly and chronically ill patients. In this work, a smart medication monitoring system is designed as a programmable IoT-based solution capable of delivering timely alerts to patients. The system employs a Raspberry Pi Pico as the central control unit, which manages both visual and audible notifications to remind patients of their scheduled medication. This not only enhances the accuracy of medicine intake but also reduces the chances of missed or delayed doses. By incorporating IoT connectivity, the system can be programmed and adapted to meet individual patient needs, ensuring flexibility and reliability in operation. The approach highlights the capability of Raspberry Pi Pico in handling control and alert functions efficiently, which directly relates to MediGuard 2.0 by demonstrating how microcontrollers can be effectively utilized for smart reminders, real-time alerts, and medication adherence in elderly care applications.

H. Daoud, M. Daoud, N. Jardak and C. Albouzidi, "Design and Implementation of Smart Pill Box for Healthcare Applications," 202320th International Multi-Conference on Systems, Signals & Devices (SSD), Mahdia, Tunisia, 2023 [3]

With the increasing demand for reliable healthcare solutions, smart pill boxes have become a promising approach to ensure timely medication intake and proper monitoring of patients. In this work, a smart pill box is designed using an Arduino Nano microcontroller integrated with an LCD, sensors, and a buzzer to provide visual and audible alerts. The system is further enhanced with mobile application support, enabling remote monitoring of the patient's medicine intake schedule. Caretakers and doctors can also receive timely updates and notifications through SMS or email in case of missed doses or irregularities, thereby ensuring continuous medical

supervision. This approach highlights the importance of Arduino Nano in handling control operations, integrating alert mechanisms, and establishing communication with external devices. The study is particularly relevant to MediGuard 2.0, as it demonstrates the effective role of microcontrollers in managing reminders, monitoring patients, and providing real-time notifications to caregivers, thereby enhancing the safety and efficiency of smart medicine dispensing systems.

V. Mishra, O. Noman, N. Bankar, A. Tiwari, S. Godbole and D. Timilsina, "Large Language Model-based Mobile Apps in OPDs:Optimized Patient Management and Prioritization," 2024 2nd DMIHER International Conference on Artificial Intelligence in Healthcare,Education and Industry (IDICAIEI), Wardha, India,2024 [4]

The integration of advanced digital tools in healthcare has opened new possibilities for improving patient care and operational efficiency. This work focuses on the use of mobile apps in outpatient departments for scheduling appointments, digital triage, and providing real-time updates to patients and healthcare providers. The system incorporates a RTC module to deliver accurate, time-based alerts, ensuring timely reminders for medication and appointments. The study emphasizes how mobile applications can enhance the overall user experience by offering better accessibility, interaction, and control for both patients and caregivers. Furthermore, the use of the RTC module (DS3231) for precise time scheduling and alerts demonstrates the potential of such components in designing reliable healthcare devices. This research is highly relevant to MediGuard 2.0, as it highlights the dual importance of mobile app integration for user-friendly control and RTC modules for dependable scheduling and reminder functions, thereby ensuring adherence to medication routines while also reducing human error, increasing efficiency in healthcare management, and supporting personalized treatment approaches for improved patient outcomes. Additionally, the combination of mobile applications and RTC modules creates a seamless ecosystem that bridges communication gaps, empowers patients with autonomy, strengthens caregiver monitoring, and paves the way for future innovations in digital health solutions, aligning with global trends in smart healthcare systems.

S. M and M. Selvi, "Smart Medical Box using Arduino," 2025 International Conference on Intelligent Computing and Control Systems (ICICCS), Erode, India, 2025 [5]

enhance the efficiency and reliability of medicine management through advanced electronics and mobile integration. One of the key features highlighted in this work is the use of a stable power supply achieved through a buck converter, which ensures that the system operates reliably without interruptions due to voltage fluctuations. The system is built around an Arduino microcontroller, which serves as the central control unit, enabling seamless operation of all components. Additionally, it incorporates GPS support, allowing users to locate nearby pharmacies when required. The project also emphasizes real-time alerts and mobile integration, which provide timely notifications to patients and caregivers regarding medicine schedules. In relevance to our project, this work provides a valuable reference for implementing effective power management using the LM2596 buck converter and integrating real-time alert systems along with voice assistance features. These aspects align closely with the objectives of our project, which aims to create a smart medicine box that ensures accurate dosage, timely reminders, and convenient monitoring for users and caregivers.

M. Abdul Kader, M. Nayim Uddin, A. Mohammad Arfi, N. Islam and M. Anisuzzaman, "Design & Implementation of an Automated Reminder Medicine Box for Old People and Hospital," 2018 International Conference on Innovations in Science, Engineering and Technology (ICISET), Chittagong, Bangladesh,2018[6]

This focuses on developing a smart medication management system tailored for elderly patients and hospital use. The system incorporates 21 airtight compartments, enabling weekly planning with three doses per day, which helps in organizing and managing multiple medications efficiently. To ensure reliable operation, the design includes voltage regulators that provide a stable and safe power supply to all electronic components. The medicine box allows flexibility in scheduling doses through either manual input or SD card-based time setup, supported by an RTC (Real-Time Clock) module for precise timing. LED indicators and speaker alerts further enhance

usability by providing visual and audio notifications, ensuring that patients do not miss their doses. In relevance to our project, the use of the AMS117 voltage regulator demonstrates an effective approach for maintaining safe and stable voltage levels, while the voice-guided instructions serve as an excellent reference for implementing elderly-friendly reminders, aligning with our goal of creating a smart medicine box that is both reliable and user-centric.

Gode, M. Narlawar, M. Thakare, N. Lakade, P. Parate and S. Kayarkar, "Smart Medicine Reminder Box," 2024 2nd International Conference on Emerging Trends in Engineering and Medical Sciences (ICETEMS), Nagpur, India, 2024 [7]

This paper presents an IoT-enabled pill box designed to improve medication adherence through programmable schedules and real-time alerts. The system features LED indicators and buzzer alerts to notify patients of their medication times, ensuring doses are not missed. Additionally, it incorporates voice-based audio alerts, enhancing accessibility for elderly or visually impaired users. In relevance to our project, this work provides guidance on implementing timely reminders and intake verification, while the DF Player Pro module serves as a useful reference for integrating voice alerts, making the medicine box more user-friendly and effective for patients.

V. Gaikwad, S. Chakrabarty, S. Jain, S. Bhat and S. Bisen, "IOT Based Medicine Box: A Step Towards An Easier Life," 2024 Second International Conference on Advances in Information Technology (ICAIT), Chikkamagaluru, Karnataka, India [8]

This paper presents an IoT-enabled medicine box designed to simplify medication management for patients while providing caregivers with real-time monitoring and control capabilities. The system employs MG996R servo motors to accurately operate the lids of individual compartments, ensuring medications are dispensed at scheduled times without errors. Visual and audio notifications are provided through LED indicators, buzzer alerts, and an LCD display, while a Real-Time Clock (RTC) module ensures precise timing for reliable operation over extended periods. The design also incorporates a dual sensory alert system, combining visual and audio

cues, and a caregiver-controlled mobile application that allows remote monitoring and management of medicine intake. In relevance to our project, this work offers valuable insights into effective compartment control using servo motors, implementation of dual alert mechanisms, and development of user-friendly applications for caregivers, all of which contribute to the creation of a smart, accessible, and dependable medicine box for patients and caregivers alike.

R. Al-Shammary, D. Mousa and S. E. Esmaeili, "The Design of a Smart Medicine Box," Electrical Engineering (ICEE), Iranian Conference on, Mashhad, Iran 2018 [9]

This paper presents a smart medicine box designed to assist elderly patients and support caregivers in managing medication schedules efficiently. The system incorporates a servo motor for precise lid control, allowing each compartment to open automatically at scheduled times. It also features a mobile application that provides SMS and email notifications to alert patients and caregivers about upcoming doses, ensuring timely medication intake. The design focuses on improving usability for elderly users while offering caregivers a reliable way to monitor adherence remotely. In relevance to our project, this work serves as a useful reference for implementing notification systems for caregivers and demonstrates the practical application of the SG90 servo motor for controlling compartment lids. These features provide guidance for creating a smart medicine box that is both accessible and functional, aligning closely with our goal of enhancing patient safety and convenience.

S. Mohammad, T. Bhowmick, M. S. U. Z. Siddique, M. M. Khan and S. Bhattacharyya, "Research and Development of a Artificial Intelligence based Smart Medicine Box," 2022 6th International Conference on Computing Methodologies and Communication (ICCMC), Erode, India, 2022 [10]

This paper discusses the development of an AI-driven smart medicine box aimed at enhancing medication management, particularly for elderly patients. The system integrates an LCD display, Arduino-based control, and an alert module to provide timely reminders for correct dosage and precise medication timing. The AI component

allows the device to intelligently manage schedules and adapt to changes in patient routines, ensuring that doses are not missed or taken incorrectly. Additionally, the system is complemented by a mobile application that enables real-time updates and scheduling, allowing caregivers to monitor patient adherence remotely. In relevance to our project, this work provides valuable insights into building an affordable and intelligent reminder system, with the Blynk-based app serving as a practical reference for implementing real-time monitoring and user-friendly control, supporting our goal of creating a smart, accessible, and reliable medicine box.

A. Mathew, J. Paul, K. Nair S., U. S. Sachin, S. Koncherry and C. V. Raghu, "Design and Implementation of a Smart Medicine Dispenser," TENCON 2019 - 2019 IEEE Region 10 Conference (TENCON), Kochi, India, 2019 [11]

This paper focuses on the development of a smart medicine dispenser aimed at improving medication adherence and ensuring accurate dosage delivery. The system is designed to automatically dispense the correct amount of medicine at preset intervals, which reduces the risk of missed or incorrect doses. It features a simple programmable interface, allowing users or caregivers to easily set and adjust medication schedules according to patient needs. In addition, the system incorporates app-based monitoring, enabling real-time updates and notifications to both patients and caregivers. In relevance to our project, this work provides valuable insights into integrating mobile applications with backend platforms like Firebase to achieve real-time alerts, automatic dispensing, and enhanced dosage accuracy. These features align closely with our objective of creating a smart medicine box that ensures timely reminders, precise medication delivery, and effective monitoring, thereby improving overall patient safety and adherence. Moreover, this study highlights the importance of combining hardware and software components to build an efficient healthcare system. The integration of app-based control with physical dispensing units allows seamless synchronization, user-friendly customization, and reliable performance. Such approaches not only reduce dependency on manual intervention but also promote long-term adherence by fostering patient engagement, caregiver confidence, and healthcare reliability in diverse medical scenarios.

B. Oluwaseun, A. Ogunmilua and E. G. AbdAllah, ”Enhancing Security of RFID Implants in the Healthcare Industry,” 2024 IEEE International Conference on RFID Technology and Applications (RFID-TA), Daytona Beach, FL, USA, 2024 [12]

This paper explores the use of RFID technology to improve patient data management and security in the healthcare industry. The system employs RFID implants to enable automatic access to patient information, particularly in emergency situations, while emphasizing secure architecture to protect sensitive data. Key features include robust privacy measures and high identification accuracy, ensuring that only authorized personnel can access patient records. In relevance to our project, this work provides a reference for implementing the RC522 RFID module to allow secure caretaker access and reliable identity verification. By incorporating such RFID-based security, our smart medicine box can ensure that only authorized users can operate the device, thereby enhancing patient safety, privacy, and controlled access to medication management.

S. Y. Zavozkin, ”Patient Health Monitoring Information System,” 2020 International Multi-Conference on Industrial Engineering and Modern Technologies (FarEastCon), Vladivostok, Russia, 2020 [13]

This paper presents a comprehensive patient health monitoring system designed to assist in rehabilitation and ongoing healthcare management. The system integrates mobile and web applications to collect and store health-related data, allowing caregivers and medical professionals to track patient progress effectively. Smartwatch devices are used to gather real-time physiological data, which is then stored in a centralized database for analysis and monitoring. In relevance to our project, this work highlights the importance of mobile and web applications in tracking and managing patient information. It also provides guidance for establishing a link between medicine intake and caregiver notifications, ensuring that patients adhere to their prescribed schedules while caregivers are kept informed, thereby enhancing the overall safety and efficiency of medication management.

B. K. Swathi Prasad, D. Sunil, A. Gowda, Kotraswamy and Vijay, "Development of Integrated Health Care App for Non - Invasive Bio - Medical Application," 2019 IEEE International Conference on Distributed Computing, VLSI, Electrical Circuits and Robotics (DISCOVER), Manipal, India, 2019 [14]

This paper discusses the development of an integrated healthcare system that focuses on non-invasive monitoring of vital health parameters such as glucose levels, pulse rate, and body mass index (BMI). The system is designed to provide real-time health data to patients and caregivers through a mobile application, which displays information in an easily understandable format. Alerts are generated for any abnormal conditions, enabling timely intervention and support. In relevance to our project, this work highlights the importance of real-time health monitoring and demonstrates effective mobile app integration, providing a framework for creating a smart medicine box that can not only manage medication schedules but also support overall patient health by delivering timely information and alerts to caregivers and patients.

S. Nizami et al., "CEA: Clinical Event Annotator mHealth Application for Real-time Patient Monitoring," 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Honolulu, HI, USA, 2018 [15]

This paper presents the development of the Clinical Event Annotator (CEA), an mHealth application designed for real-time patient monitoring. The system operates on both Android and web platforms, allowing caregivers and healthcare professionals to track patient activities and events efficiently. Key features include real-time event annotation, which records alarms, patient care activities, and movements, generating detailed reports and training data for further analysis. In relevance to our project, this work validates the use of mobile health applications for continuous patient monitoring, highlighting the importance of caregiver alerts, data recording, and real-time supervision. These functionalities provide valuable insights for implementing a smart medicine box that not only delivers timely medication reminders but also allows caregivers to continuously monitor patient adherence and well-being.

W. M. F. W. M. Rosdi, S. A. Suhaimi, N. A. Md Lazam, A. J. Alias, F. Abdullah and S. N. Azemi, "Smart Pill Dispenser with Monitoring System," 2021 IEEE Symposium on Wireless Technology & Applications (ISWTA), Shah Alam, Malaysia, 2021 [16]

This paper presents an IoT-based smart pill dispenser designed to automate medication management and enhance patient adherence. The system integrates a mobile application for scheduling, real-time alerts, and caregiver notifications, ensuring that patients take their medication on time. Cloud-based monitoring, implemented through platforms like Firebase, allows data to be stored securely and accessed remotely, providing caregivers with continuous updates on patient adherence. In relevance to our project, this work reinforces the concept of automatic dispensing and scheduling, validates the use of real-time alerts and caregiver updates, and serves as inspiration for designing a secure, portable, and user-friendly smart medicine box that combines IoT technology with effective monitoring and communication features.

A. N. Raj, G. S. Kumar, V. Srinivasan, D. V. L. Reddy, B. V. F. Justin and R. Priyadharshini, "Analysis of Internet of Things based Medication Dispenser Reminder and Drug Expiration Health Monitoring and its Applications," 2024 4th International Conference on Sustainable Expert Systems (ICSES), Kaski, Nepal, 2024 [17]

This paper explores the design and implementation of a smart medicine box equipped with IoT sensors to manage medication schedules and monitor drug expiration. The system integrates Wi-Fi connectivity and a mobile application to provide remote updates, enabling caregivers to track medication intake and ensure timely reminders for patients. Alerts are generated not only for scheduled medicine times but also for approaching drug expiry dates, enhancing safety and adherence. In relevance to our project, this work provides valuable guidance on implementing real-time reminders and app-based monitoring, supporting the development of a smart medicine box that can effectively manage medication schedules while keeping caregivers informed, ensuring both convenience and patient safety.

P. A. Harsha Vardhini, M. S. Harsha, P. N. Sai and P. Srikanth, "IoT based Smart Medicine Assistive System for Memory Impairment Patient," 2020 12th International Conference on Computational Intelligence and Communication Networks (CICN), Bhimtal, India, 2020 [18]

This paper presents an IoT-based smart medicine assistive system specifically designed for patients with memory impairments, focusing on ensuring timely medication intake. The system tracks medicine schedules and provides alerts to patients at the correct times, reducing the risk of missed doses and improving overall adherence. The design emphasizes affordability and ease of use, making it particularly suitable for elderly patients and individuals who may be illiterate or have difficulty operating complex devices. In relevance to our project, this work inspires the development of the IoT reminder system in MediGuard 2.0, highlighting the importance of creating a user-friendly, elderly-centric, and cost-effective smart medicine box that supports accurate medication management and improves patient safety.

N. Chandana, M. Kumar, Megha, G. S. Hegde and K. Shashi Raj, "IoT based Medicine Reminder and Dispensing Machine," 2022 Second International Conference on Artificial Intelligence and Smart Energy (ICAIS), Coimbatore, India, 2022 [19]

This paper discusses the development of an IoT-based smart medicine dispenser designed to automate medication management and enhance patient adherence. The system stores and schedules weekly doses, allowing patients to receive the correct medication at the right time without manual intervention. An SD card is used to track medicine intake, enabling doctors or caregivers to review patient compliance and maintain accurate records. Key features include automatic reminders through both app notifications and buzzer alerts, ensuring timely medication intake. In relevance to our project, this work supports the implementation of reminder functions and demonstrates how pill scheduling can be automated, providing practical guidance for developing a smart medicine box that is both efficient and user-friendly, improving overall patient safety and adherence.

B. -b. Chen, Y. -h. Ma and J. -L. Xu, "Research and Implementation of an Intelligent Medicine Box," 2019 4th International Conference on Intelligent Green Building and Smart Grid (IGBSG), Hubei, China, 2019 [20]

This paper presents the design and implementation of an intelligent medicine box that combines sensors, motor mechanisms, and mobile application interfaces to improve medication management. The system uses a TM32 microcontroller as the central control unit, coordinating various functions including the detection of human presence through sensors. This feature ensures that the medicine box operates only when a user is present, enhancing safety and reducing misuse. The device also employs motor-controlled rotation to dispense medicines accurately from designated compartments, improving convenience and dosage reliability. Additionally, a mobile application with a user-friendly interface allows caregivers and patients to monitor and manage medicine schedules effectively. In relevance to our project, this work provides valuable insights into integrating sensors, motors, and app interfaces, guiding the development of MediGuard 2.0 to be both intelligent and accessible while maintaining precise control over medicine dispensing.

M., Nirmala Devi, et al. "An Automated Drug Dispenser Mobile Application Using Internet of Things (IoT) in Smart Healthcare Systems." Technologies for Sustainable Healthcare Development, edited by Thangavel Murugan, et al., IGI Global Scientific Publishing, 2024 [21]

This paper presents the development of an automated drug dispenser system that integrates IoT technology with a mobile application to enhance smart healthcare management. The system features a centralized cloud database, which ensures that patient information, medication schedules, and dispensing data are securely stored and updated in real-time. The design also incorporates patient identity verification to ensure that only authorized users can access the medication, enhancing safety and reliability. In relevance to our project, this work demonstrates how combining IoT with mobile applications can improve medication management, patient safety, and user experience. It provides valuable guidance for designing MediGuard 2.0 to include

real-time updates, secure access, and effective caregiver monitoring, ensuring both convenience and reliability in medication adherence.

Philip, J., Abraham, F.M., Giboy, K.K., Feslina, B.J., and Rajan, T. "Automatic Medicine Dispenser Using IoT." International Journal of Engineering Research and Technology (IJERT), vol. 9, no. 8, 2020 [22]

This paper presents the design and implementation of an IoT-based automatic medicine dispenser aimed at improving medication adherence over extended periods. The system is capable of storing and dispensing medicines for 2–3 weeks, reducing the need for frequent manual intervention and ensuring continuity in patient care. It integrates a mobile application that allows caregivers and patients to control and monitor the dispenser remotely, providing timely updates on medication schedules. Notifications are generated for missed doses, helping to prevent skipped or delayed medication intake. In relevance to our project, this work demonstrates how IoT-enabled systems can enhance timely medicine consumption, support remote monitoring, and improve overall patient adherence. These insights are useful for guiding the development of MediGuard 2.0, ensuring it is both efficient and user-friendly for patients and caregivers.

J., Ramkumar, C., Karthikeyan, E., Vamsidhar, and K.N., Dattatraya. "Automated Pill Dispenser Application Based on IoT for Patient Medication." IoT and ICT for Healthcare Applications, edited by N. Gupta and S. Paiva, 2020 [23]

This paper presents the development of an IoT-based automated pill dispenser designed to ensure proper medication intake and improve patient adherence. The system enables interaction between patients and doctors through IoT connectivity, allowing caregivers to monitor adherence and make necessary adjustments remotely. Cloud integration facilitates secure storage of patient data, real-time monitoring, and precise dosage control, enhancing both safety and reliability. Additionally, the paper provides a comparative analysis of different pill dispensers, highlighting the advantages of IoT-enabled systems. In relevance to our project, this work emphasizes how IoT and cloud technologies can improve pill intake monitoring, ensure accurate

dosage, and enhance overall healthcare efficiency. These insights are valuable for guiding the design of MediGuard 2.0, enabling it to provide timely reminders, remote monitoring, and accurate medication management.

S., Sharmin, M.I.K.U., Ratan, and A.H., Piash. "An Automated and Online-Based Medicine Reminder and Dispenser." Proceedings of the International Conference on Big Data, IoT, and Machine Learning. Lecture Notes on Data Engineering and Communications Technologies, vol. 95, edited by M.S. Arefin, M.S. Kaiser, A. Bandyopadhyay, M.A.R. Ahad, and K. Ray, Springer, Singapore, 2022. [24]

This paper presents the development of an automated medicine reminder and dispenser system that integrates IoT and online-based functionalities to enhance patient medication adherence. The system features an LCD display, buzzer, and LED indicators to provide timely reminders for medication intake, ensuring that doses are not missed. It includes an input interface that allows patients or caregivers to add or modify prescriptions easily and an output interface that displays reminders and logs medication history for future reference. The system achieved a 90 percentage success rate and high user satisfaction, demonstrating its effectiveness in real-world scenarios. In relevance to our project, this work provides valuable guidance on designing reminders using IoT and mobile app integration, showing how such systems can significantly improve medication adherence, user awareness, and overall healthcare management.

P. S V and S. M, "IoT Enabled Smart Medicine Box for Pharmaceutical Health care Assistance," 2025 5th International Conference on Soft Computing for Security Applications (ICSCSA), Salem, India, 2025 [25]

This paper presents the design of an IoT-enabled smart medicine box developed to enhance medication management, particularly for elderly patients and individuals suffering from chronic conditions who require strict adherence to treatment schedules. The system is integrated with an Android mobile application that plays a central role by providing timely reminders and alerts for medicine intake, thereby minimizing

the chances of missed or incorrect doses. In addition to reminders, the system is designed to track critical information such as dosage frequency, timing, and remaining medicine stock, ensuring that both patients and caregivers are constantly updated with real-time data. Such integration of IoT and mobile platforms not only makes the system user-friendly but also reduces dependency on memory or manual tracking, which is often difficult for elderly users. By allowing caregivers to monitor the medication process remotely and by automating essential tasks, the smart medicine box effectively reduces caregiver workload while simultaneously improving patient safety and adherence. In relevance to our project, this work provides an important demonstration of how IoT-based systems combined with mobile applications can significantly enhance medication adherence, improve convenience for patients, and create a reliable framework that supports caregivers, thereby influencing the design and goals of our MediGuard 2.0 solution.

P. R, M. R. Karaguppi, R. S. R and G. S. S, "Medibot: An AI-Enabled IoT-Based Smart Pill Dispenser with App-Based Dosage and Expiration Alerts," 2025 3rd International Conference on Smart Systems for applications in Electrical Sciences (ICSSES), Tumakuru, India, 2025 [26]

This paper presents the design and development of Medibot, an AI- and IoT-enabled smart pill dispenser aimed at improving medication adherence, safety, and caregiver support. The system integrates voice recognition to allow patients to interact with the device easily, as well as location-based delivery features for added convenience. A mobile application is used for scheduling, real-time monitoring, and sending alerts, ensuring that patients take their medications on time. The device also incorporates biometric security to prevent unauthorized access, and includes features for low-stock and expiry monitoring to maintain accurate and safe medication management. In relevance to our project, this work demonstrates how the integration of AI and IoT technologies can enhance the functionality, safety, and reliability of smart medicine boxes, providing valuable guidance for developing MediGuard 2.0 with intelligent reminders, secure access, and comprehensive caregiver support, ultimately improving patient independence, healthcare efficiency, and long-term treatment outcomes.

V. Gaikwad, S. Chakrabarty, S. Jain, S. Bhat and S. Bisen, "IOT Based Medicine Box: A Step Towards An Easier Life," 2024 Second International Conference on Advances in Information Technology (ICAIT), Chikkamagaluru, Karnataka, India, 2024 [27]

This paper presents an IoT-enabled medicine box designed to simplify medication management while providing real-time monitoring for caregivers. The system utilizes the Blynk mobile application to send alerts to caregivers, ensuring timely intervention in case of missed doses. An ultrasonic sensor is employed to detect patient intake, confirming whether the medicine has been taken, while a load cell monitors pill quantity and provides refill reminders when supplies are low. In relevance to our project, this work validates the use of real-time mobile alerts and demonstrates how integrating sensors for intake detection and refill management can enhance the efficiency and reliability of a smart medicine box. These insights are particularly useful for designing MediGuard 2.0 to be both intelligent and user-friendly for patients and caregivers.

T. Zhang, Z. Liu, X. Cao and W. Xia, "Intelligent Medicine Box Monitoring and Management System," 2015 International Conference on Computer Science and Mechanical Automation (CSMA), Hangzhou, China, 2015 [28]

This paper presents an intelligent medicine box system designed to enhance medication management and patient adherence. The system incorporates RFID technology for secure medicine identification, ensuring that the correct medication is dispensed to the patient. A touch screen display provides detailed information including medicine name, dosage, and timing, improving usability and reducing errors. The device is network-enabled, allowing caregivers to monitor and control medicine schedules remotely, while patient reminders ensure timely intake. In relevance to our project, this work strengthens the concept of RFID-based verification and highlights how such features can help prevent wrong or missed doses. These insights are valuable for designing MediGuard 2.0 to be secure, efficient, and reliable, ensuring accurate medication delivery and improved patient safety. Additionally, the study

emphasizes the integration of real-time alerts, automated logging of medicine intake, and secure data transmission to caregivers, which not only enhances accountability but also supports personalized healthcare. The combination of interactive interfaces, RFID security, and connectivity provides a comprehensive framework that can guide the development of advanced smart medicine boxes. Implementing similar features in MediGuard 2.0 can promote better adherence, reduce human errors, enable timely interventions, and ultimately improve overall healthcare outcomes for patients with complex medication routines. Furthermore, the research highlights the importance of user-centric design, demonstrating that intuitive interfaces and clear feedback mechanisms significantly enhance patient engagement. The integration of cloud-based platforms allows seamless synchronization between devices, caregivers, and healthcare providers, enabling analytics-driven insights for proactive care. By incorporating such multi-layered security, real-time monitoring, and automated control, MediGuard 2.0 can ensure consistent compliance, facilitate data-driven decision-making, and support long-term management of chronic conditions, ultimately fostering a safer, more efficient, and reliable medication management system.

K. S. Divya, A. Raguraj, G. Devanand, A. Thomas, G. P. Mannickathan and N. K. Nataraj, "Automatic Pill Dispenser for Elderly Medical Adherence: A Survey," 2024 IEEE International Conference on Computing, Power and Communication Technologies (IC2PCT), Greater Noida, India, 2024 [29]

This paper presents a comprehensive survey of various automatic pill dispenser systems, with particular emphasis on IoT-based medicine dispensers that are increasingly developed to support elderly medical adherence. It highlights their applications in both home environments and nursing shelters, emphasizing how strict compliance with prescribed medication schedules plays a crucial role in improving patient health, reducing complications, and easing caregiver responsibilities. The survey reviews the technological design, working mechanisms, and practical implementations of different existing systems, while also analyzing their strengths and weaknesses in terms of accessibility, reliability, cost-effectiveness, and overall user-friendliness. It points out the major limitations of earlier solutions, such as difficulties faced by elderly users in

handling complex designs, limited integration with real-time monitoring, and gaps in ensuring continuous adherence. In direct relevance to our project, this work provides valuable insights into previous attempts and their shortcomings, which in turn guide the design of MediGuard 2.0, a more effective and IoT-enabled smart medicine box that seeks to overcome these issues by enhancing ease of use, ensuring reliable dispensing, enabling continuous monitoring, and providing caregiver support to achieve better adherence and overall healthcare management for elderly patients.

K. R. Karthikeyan, E. Dharan Babu, S. Ranjith and S. Arunkumar, "Smart Pill Dispenser for Aged Patients," 2021 International Conference on Advancements in Electrical, Electronics, Communication, Computing and Automation (ICAECA), Coimbatore, India, 2021 [30]

This paper presents the design of a smart pill dispenser specifically aimed at elderly patients, providing autonomous medicine dispensing based on a programmed schedule. The system tracks medicine usage through cloud storage, enabling caregivers to monitor adherence remotely and maintain accurate records. Additionally, the device is controlled and monitored via a mobile application, providing real-time updates and notifications to ensure that patients take their medications on time. In relevance to our project, this work inspires the integration of automated dispensing, scheduled medicine management, and mobile monitoring in MediGuard 2.0, supporting the creation of a reliable, user-friendly, and intelligent system for elderly patient care. Furthermore, the study emphasizes the importance of combining hardware automation with software connectivity, ensuring both accurate medicine delivery and seamless caregiver communication. The system's cloud integration allows data analytics on patient adherence patterns, which can help in adjusting treatment plans and preventing complications. By incorporating reminders, alerts, and usage logs, the device enhances accountability, reduces human error, and promotes patient independence. These insights are particularly valuable for designing MediGuard 2.0 to be scalable, secure, and adaptable to varying medication routines, ultimately improving long-term health outcomes and providing peace of mind to both patients and caregivers.

Chapter 3

MEDIGUARD 2.0 - A SMART, SECURE MEDICINE BIX WITH APP MONITORING

3.1 Motivation

The motivation for this project arises from the urgent need for a reliable, secure, and intelligent medication management system that can address the challenges faced by elderly individuals, patients with chronic diseases, and those with memory-related conditions. In today's healthcare scenario, forgetting to take medicines, taking the wrong dosage, or consuming them at the wrong time are common issues that can lead to serious health complications, prolonged treatments, hospitalization, or even life-threatening emergencies. Moreover, in many households, medicines are stored in simple boxes or containers without proper security, leaving them vulnerable to unauthorized or accidental access—particularly by children or unintended users—thus creating significant safety risks. To tackle these problems, the system integrates advanced features such as biometric fingerprint access and RFID authentication, ensuring that only authorized users or caregivers can open the device. A stepper motor-driven rotating tray organizes medicines into multiple compartments, enabling precise dispensing based on the scheduled time. Patients are guided through OLED display messages and voice alerts, ensuring clarity and ease of use, while an infrared sensor

confirms whether the medicine has been picked up. Additionally, a companion mobile application allows caregivers to configure schedules, monitor adherence, and receive real-time notifications, reducing the burden of constant supervision. By merging automation, security, real-time monitoring, and caregiver connectivity, this project aims to create a safe, user-friendly, and efficient solution that not only minimizes medication errors but also enhances the independence, confidence, and overall quality of life of patients.

3.2 Objectives

The objective of this project is to develop an advanced smart medication management system that assists elderly individuals, patients with chronic illnesses, and those with special needs in adhering to their prescribed medicine schedules accurately and securely. By integrating modern technologies, the system ensures timely medication intake, reduces the risk of errors, and enhances patient safety. The device incorporates a rotating tray with eight compartments powered by a stepper motor for precise dispensing, along with biometric fingerprint access for patients and RFID authentication for caregivers to restrict unauthorized usage. An OLED display provides real-time instructions, while voice alerts remind users of medication times, ensuring clarity and accessibility for all. An infrared sensor detects whether medicines are picked up, offering an added layer of verification. The project emphasizes automation, security, and user-friendliness, supporting independent medication adherence with minimal caregiver intervention. Its secure access mechanisms prevent accidental or premature handling of medicines, thereby increasing reliability and safety. Furthermore, the companion mobile application enables caregivers to schedule medicines, receive real-time notifications, and monitor adherence remotely, ensuring continuous supervision without physical presence. Beyond improving medication compliance, the system also reduces confusion and stress, ultimately contributing to a better quality of life for patients. By focusing on affordability, practicality, and ease of use, the project aims to provide an effective and widely adaptable solution for home healthcare and assisted living environments.

3.3 Methodology

1. Eight-Compartment Rotating Tray: The medicine box consists of a rotating tray divided into eight compartments, each designated for storing a different type of medicine. A stepper motor is used to rotate the tray with accurate position indexing so that the required compartment is aligned and accessible at the scheduled time. This structured mechanism ensures precision in dispensing and minimizes errors.

2. Fingerprint and RFID-Based Access Control: The system uses a fingerprint sensor for patients and dual authentication (fingerprint + RFID) for caregivers to ensure secure access. Patients can open the device using their fingerprint, while caregivers use both RFID and fingerprint during refilling. This dual verification ensures medicines remain tamper-proof and secure.

3. Voice Alerts and OLED Display: At scheduled times, the system provides reminders using both voice alerts and an OLED display. Voice instructions guide the patient to take the correct medicine, while the OLED shows details such as medicine type, dosage, and timing. This improves usability, especially for elderly patients or those with memory issues.

4. IR Sensor for Medicine Pickup and Auto-Shutdown: An infrared (IR) sensor detects the hand motion when the patient picks up the medicine. Once the intake is detected, the system records the action and ensures adherence is logged. After the medicine is taken, the box automatically shuts down within 3 seconds, preventing misuse, accidental access, or prolonged exposure of the medicine.

5. Refill Mode for Caregivers: The box provides a refill mode that can be accessed only by caregivers through RFID and fingerprint verification. Once authenticated, the box unlocks for a fixed duration, allowing secure refilling of compartments. After this period, the system locks automatically, preventing unauthorized use.

6. Mobile Application Integration: The system is integrated with a mobile app that manages scheduling, provides real-time alerts, and updates caregivers about patient adherence. Caregivers can monitor intake records, receive notifications for missed doses, and track refill requirements remotely.

7. Rechargeable Power Backup: The device is powered by a rechargeable lithium-ion battery pack. This ensures uninterrupted operation even during power

failures, making the system portable, reliable, and suitable for both home and healthcare settings.

3.4 Block diagram

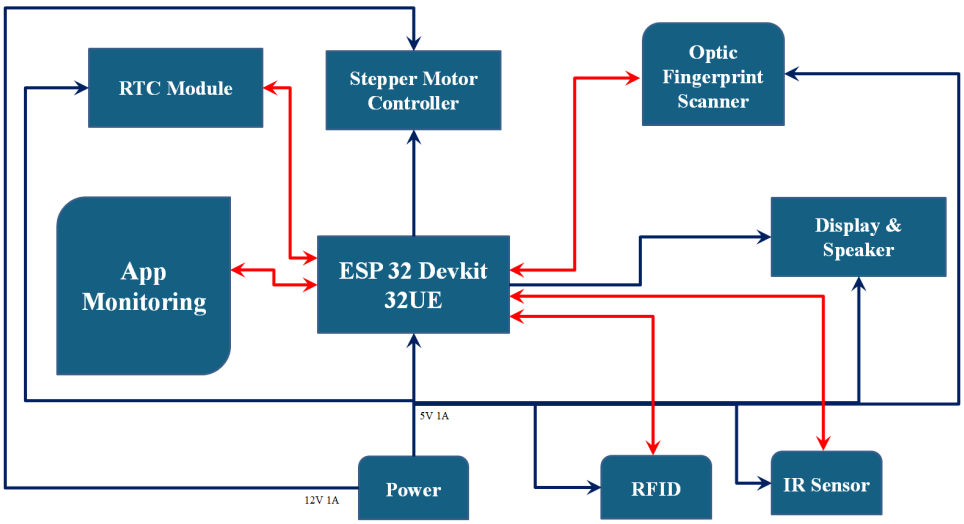


Figure 3.1: Block diagram of MediGuard 2.0 System

The block diagram of MediGuard 2.0 represents an embedded medication management system controlled by the ESP32 DevKitC-32UE microcontroller, which coordinates all modules for secure and automated operation. User authentication is achieved through the R307 fingerprint scanner for patients and the RC522 RFID reader for caregivers, ensuring restricted and role-specific access. A stepper motor, driven by the DRV8825 driver, rotates the eight-compartment tray to align the correct slot based on the scheduled time. Accurate scheduling is maintained using a DS3231 RTC module with internet time synchronisation for reliability. For patient interaction, a 0.96-inch OLED display shows instructions and status updates, while a DFPlayer Mini audio module with PAM8403 amplifier and speaker provides voice-based alerts and reminders. An infrared sensor is placed at the dispensing area to detect whether medicine has been picked up, offering an additional layer of confirmation. The entire system is powered by a 3-cell lithium-ion battery pack with a 3S BMS protection circuit and regulated through an LM2596 buck converter. Additionally, the system connects to a mobile application, enabling caregivers to configure schedules, monitor

adherence, receive real-time notifications, and manage language preferences, thereby combining automation, security, and connectivity into a reliable solution for safe healthcare management.

3.5 Circuit Diagram

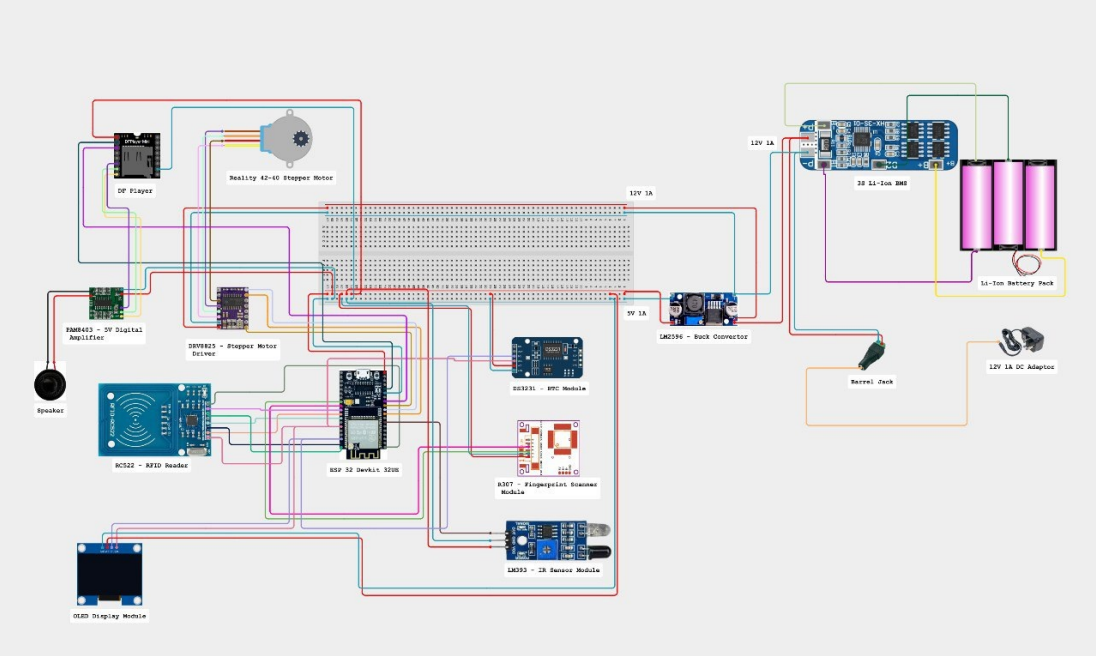


Figure 3.2: Circuit Diagram of Mediguard 2.0

MediGuard 2.0 is designed as a smart medication management and alert system that operates under the control of the ESP32 DevKitC 32UE microcontroller, which serves as the central processing and communication unit. The system is developed to address the challenges of safe and timely medication intake by combining automation, biometric security, and real time monitoring. User authentication is managed through an R307 fingerprint Reader connected to the ESP32 via the UART2 interface, ensuring that only the authorized patient can access the device. To provide caregiver level access, an RC522 RFID reader is interfaced with the ESP32 through the SPI protocol, allowing caregivers to refill or configure the device securely. Scheduling is handled by a DS3231 real time clock module connected to the I2C bus, ensuring precise timing for medication alerts even during network unavailability. Additionally, the ESP32 can synchronize with internet based time services whenever Wi Fi connectivity

is available, providing highly reliable scheduling and consistency. The dispensing mechanism of MediGuard 2.0 is based on a rotating tray with eight compartments, driven by a Creality 42 40 stepper motor and controlled by a DRV8825 stepper driver. The driver is powered by the twelve-volt supply from the battery pack, with control signals such as step, direction, and enable lines coming directly from the ESP32 GPIO pins. A stabilising capacitor is placed across the motor power supply to prevent voltage fluctuations during operation. The tray rotates precisely to present the correct compartment at the scheduled time, while an infrared sensor module connected to a digital input pin verifies whether the patient has taken the medicine, thereby adding accountability and improving adherence tracking. For user interaction, a 0.96 inch OLED display connected via I2C presents real time information, medication schedules, and instructions, while a DFPlayer Mini module connected via UART1 works together with a PAM8403 amplifier and a compact speaker to provide voice alerts and reminders, ensuring accessibility for elderly and visually impaired patients. The power system is built around a three cell lithium ion battery pack connected in series and managed by a three series battery management system module, which provides protection against overcharging, over discharging, and short circuits. The battery supplies approximately twelve volts, which is regulated down to five volts by an LM2596 buck converter to power the ESP32 and peripheral modules such as the fingerprint scanner, RFID reader, display, audio system, and sensors. A 12V charging interface is incorporated to make recharging the battery pack simple and reliable, while uninterrupted operation is ensured even during power cuts. All components are connected with a common ground to maintain circuit stability and prevent electrical noise or fluctuations. During real time operation, the ESP32 continuously monitors the DS3231 clock for scheduled intervals, authenticates users through fingerprint or RFID, rotates the tray to the correct medicine compartment, triggers audio and visual alerts, and confirms pickup using the infrared sensor. Through its Wi Fi connectivity, it also links to a mobile application that allows caregivers to set schedules, monitor adherence, and receive immediate notifications in case of missed doses. By integrating automation, biometric security, voice guidance, and app-based monitoring, MediGuard 2.0 provides a detailed, intelligent, and reliable solution for modern medication management in home and assisted living environments.

3.6 Components used

3.6.1 ESP32 DevKitC 32UE

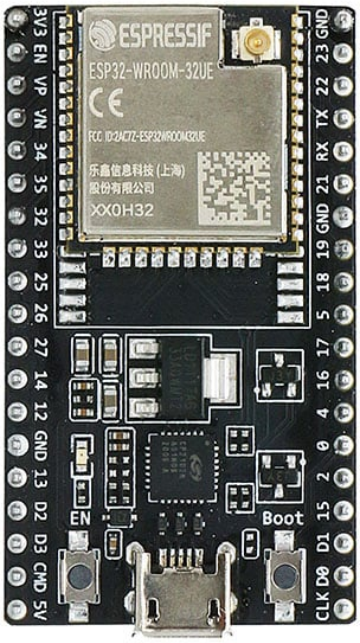


Figure 3.3: ESP32 DevKitC 32UE

The ESP32 DevKitC 32UE is a Wi-Fi and Bluetooth-enabled microcontroller development board widely used in IoT and automation applications. It has a compact design with dimensions of 54 × 27 mm. The board operates within a voltage range of 3.0V to 3.6V (5V input via USB) and is powered by a dual-core 32-bit LX6 processor running up to 240 MHz. The ESP32 DevKitC 32UE provides 520 KB SRAM with external flash support up to 16 MB and features 38 GPIO pins configurable for ADC, DAC, PWM, I2C, SPI, UART, CAN, and capacitive touch inputs. It supports 2.4 GHz Wi-Fi (802.11 b/g/n) and Bluetooth v4.2 BR/EDR + BLE with an external U.FL antenna connector for enhanced range. The board also integrates a USB-to-serial interface for easy programming and debugging. Its low-power modes make it suitable for battery-operated devices, ensuring high performance and reliability in IoT devices, smart homes, robotics, and industrial automation applications.

3.6.2 FINGERPRINT READER (R307)



Figure 3.4: FINGERPRINT READER (R307)

The Fingerprint Reader R307 is an optical fingerprint recognition module widely used in IoT, smart locks, attendance systems, and secure authentication applications. It operates at a voltage range of 3.6V to 6V with a typical operating current of 80 mA. The module features a high-resolution 256 × 288-pixel optical sensor for capturing fingerprint images and has an onboard processor for template storage and matching, supporting up to 1,000 fingerprint templates internally. It communicates via UART interface with adjustable baud rates, making it easy to integrate with microcontrollers such as Arduino, ESP32, and Raspberry Pi. With a fast recognition speed of less than 1 second and high reliability, the R307 is suitable for real-time security applications, access control, and personal identification systems.

3.6.3 IR SENSOR (LM393)

The IR Sensor LM393 is an infrared obstacle detection module widely used in line-following robots, object detection, and automation projects. It operates at a voltage range of 3.3V to 5V and has a typical current consumption of 20 mA. The module consists of an infrared transmitter and receiver pair along with the LM393 comparator for digital output. It provides both digital and analog output, allowing easy interfacing with microcontrollers such as Arduino, ESP32, and Raspberry Pi. The detection range is adjustable via an onboard potentiometer, typically between 2 cm to 30 cm. The

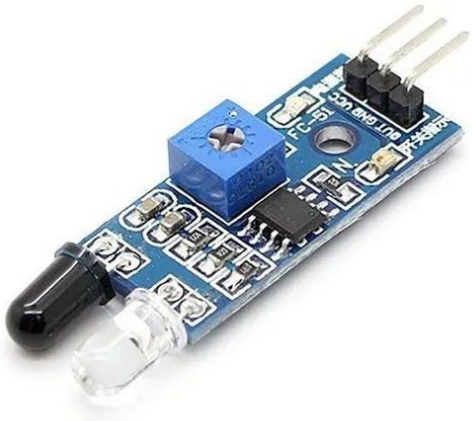


Figure 3.5: IR SENSOR (LM393)

LM393 sensor is fast, reliable, and suitable for real-time detection, obstacle avoidance, and automation applications.

3.6.4 RFID MODULE (RC522)



Figure 3.6: RFID MODULE (RC522)

The RFID Module RC522 is a highly integrated reader/writer module widely used in access control, inventory management, and IoT applications. It operates at a voltage range of 3.3V to 3.6V with a typical current consumption of 13–26 mA during reading. The module supports communication with microcontrollers via the SPI interface, and it is compatible with Arduino, ESP32, and Raspberry Pi platforms. It can read and write data on 13.56 MHz RFID cards and tags with a typical read

distance of 2–5 cm. The RC522 module features onboard antennas, high reliability, and fast response, making it suitable for secure authentication, attendance systems, and contactless identification applications.

3.6.5 STEPPER MOTOR (CREALITY 42-40)



Figure 3.7: STEPPER MOTOR (CREALITY 42-40)

The Creality 42-40 is a bipolar stepper motor widely used in robotics, CNC machines, and automation projects. It operates at a voltage range of 12V to 24V with a typical current of 1.5 A per phase. The motor has a step angle of 1.8° (200 steps per revolution), providing precise positioning and control. It features high torque, low vibration, and reliable performance, making it suitable for applications requiring accurate movement. The Creality 42-40 stepper motor is compatible with stepper drivers and microcontrollers such as Arduino, ESP32, and Raspberry Pi for easy integration in automated systems.

3.6.6 STEPPER MOTOR DRIVER MODULE (DRV8825)

The DRV8825 is a high-performance stepper motor driver module widely used in CNC machines, 3D printers, and robotics for precise motor control. It operates at a voltage

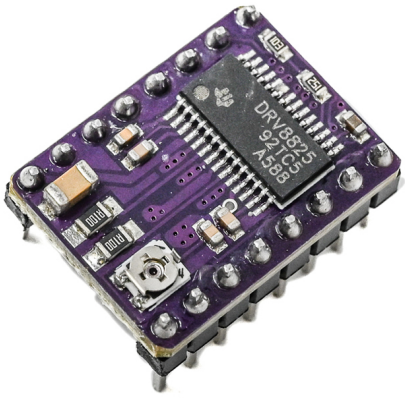


Figure 3.8: STEPPER MOTOR DRIVER MODULE (DRV8825)

range of 8.2V to 45V and supports a peak current of up to 2.5 A per phase with proper cooling. The module features adjustable current limiting, over-temperature shutdown, and five microstepping resolutions (full, half, quarter, eighth, and sixteenth steps) for smooth and accurate motor operation. It is compatible with bipolar stepper motors and can be easily interfaced with microcontrollers such as Arduino, ESP32, and Raspberry Pi, making it ideal for automated and precise motion control applications.

3.6.7 Mini DF PLAYER

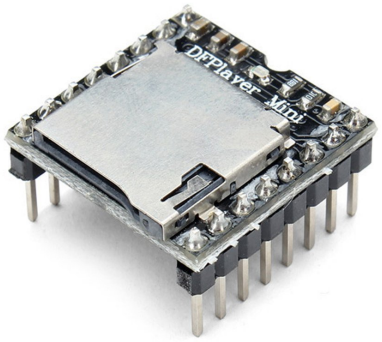


Figure 3.9: Mini DF PLAYER

The Mini DF Player is a compact MP3 audio player module widely used in embedded systems, IoT devices, and automation projects for audio playback. It

operates at a voltage range of 3.2V to 5V with a typical current consumption of 20–100 mA during playback. The module supports MP3, WAV, and WMA audio formats from microSD cards and features an onboard DAC for direct speaker output. It communicates via UART or serial interface, making it easy to interface with microcontrollers such as Arduino, ESP32, and Raspberry Pi. Its small size, low power consumption, and simple control commands make it suitable for voice alerts, audio notifications, and interactive projects.

3.6.8 MINI 5V DIGITAL AMPLIFIER (PAM8403)

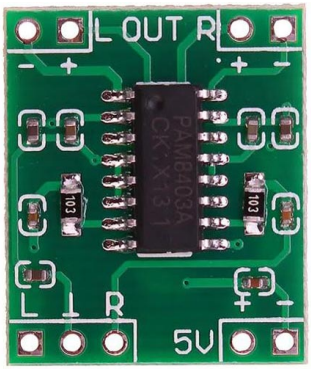


Figure 3.10: MINI 5V DIGITAL AMPLIFIER (PAM8403)

The PAM8403 is a compact 3W+3W stereo digital audio amplifier module widely used in portable audio, IoT, and embedded projects. It operates at a voltage range of 2.5V to 5.5V and provides high-efficiency amplification with low heat generation. The module features stereo output, over-current protection, and low total harmonic distortion (THD+N ¡0.1 percentage). It can be easily interfaced with microcontrollers such as Arduino, ESP32, and Raspberry Pi, making it ideal for small speaker systems, audio notifications, and portable audio applications.

3.6.9 0.5W SPEAKER (2.5”)

The 0.5W 2.5” speaker is a compact audio output device widely used in embedded systems, IoT devices, and small electronic projects. It operates at a voltage range of 3V to 5V and provides a power output of 0.5 watts. The speaker features a 2.5-inch diaphragm for clear sound reproduction and is compatible with microcontrollers and



Figure 3.11: 0.5W SPEAKER (2.5”)

audio modules such as Arduino, ESP32, DF Player, and PAM8403 amplifier. Its small size and low power consumption make it suitable for voice alerts, notifications, and small audio playback applications.

3.6.10 OLED DISPLAY MODULE



Figure 3.12: OLED DISPLAY MODULE

The OLED Display Module is a compact, low-power display widely used in IoT devices, embedded systems, and microcontroller projects. It operates at a voltage range

of 3.3V to 5V and features a resolution of 128 × 64 pixels for clear and sharp text and graphics. The module supports I2C and SPI communication interfaces, making it easy to interface with microcontrollers such as Arduino, ESP32, and Raspberry Pi. Its small size, high contrast, and low power consumption make it ideal for displaying sensor data, status information, and interactive menus in compact electronic projects.

3.6.11 LI-ION BATTERY



Figure 3.13: LI-ION BATTERY

The Li-ion battery is a rechargeable energy storage device widely used in portable electronics, IoT devices, and embedded systems. It typically operates at a nominal voltage of 3.7V with capacities ranging from 500 mAh to 3000 mAh, depending on the application. The battery features high energy density, low self-discharge, and long cycle life, making it suitable for battery-operated projects. It can be easily integrated with microcontrollers and modules such as ESP32, sensors, and displays, providing reliable power for portable and mobile electronic applications.

3.6.12 3S LI-ION BMS

The 3S Li-ion BMS (Battery Management System) is a protective circuit module designed for 3-cell (3S) lithium-ion battery packs. It operates at a nominal voltage of 11.1V (3 × 3.7V) and provides overcharge, over-discharge, overcurrent, and short-circuit protection to ensure safe operation. The module balances the charge across all three cells, extending battery life and maintaining optimal performance. It is widely used in portable electronics, IoT devices, and DIY battery pack applications.

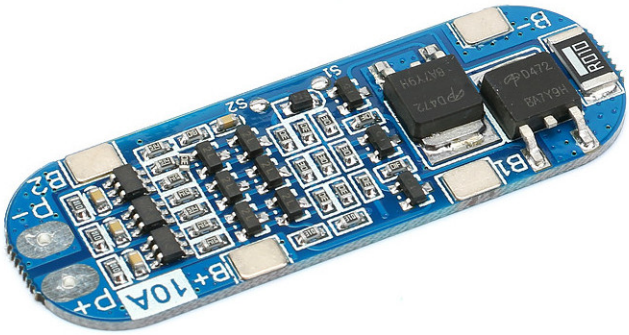


Figure 3.14: 3S LI-ION BMS

3.6.13 12V 1A DC ADAPTER

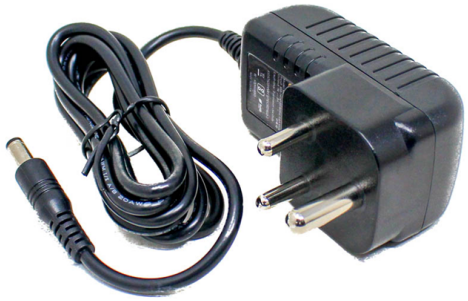


Figure 3.15: 12V 1A DC ADAPTER

The 12V 1A DC adapter is a reliable DC power supply widely used to provide stable voltage for microcontrollers, sensors, and electronic modules. It operates with an input voltage range of 100–240V AC and delivers a regulated 12V DC output with a maximum current of 1A. The adapter includes built-in overcurrent, overvoltage, and short-circuit protection, ensuring safe and consistent operation for connected devices. Its compact design and high efficiency make it suitable for IoT projects, automation systems, embedded electronics, and portable devices requiring a steady 12V power source.

3.6.14 BUCK CONVERTER (LM2596)



Figure 3.16: BUCK CONVERTER (LM2596)

The LM2596 Buck Converter is a voltage step-down module widely used in power supply regulation for microcontrollers, sensors, and electronic projects. It operates with an input voltage range of 4.5V to 40V DC and provides a stable adjustable output voltage from 1.25V to 37V DC with a maximum output current of 3A. The module features overcurrent, overtemperature, and short-circuit protection, ensuring safe and reliable operation. Its compact design, high efficiency, and easy adjustability make it suitable for IoT devices, automation systems, and embedded electronics requiring regulated DC voltage.

3.6.15 RTC MODULE (DS3231)



Figure 3.17: RTC MODULE (DS3231)

The DS3231 RTC (Real-Time Clock) module is a highly accurate timekeeping device widely used in IoT, embedded systems, and automation projects. It operates at a voltage range of 3.3V to 5V and maintains time in hours, minutes, and seconds, along with day, date, month, and year, including leap year correction. The module features an onboard temperature-compensated crystal oscillator (TCXO) for precise timekeeping

and a backup battery input to retain time during power outages. It communicates via the I2C interface, making it easy to interface with microcontrollers such as Arduino, ESP32, and Raspberry Pi. Its high accuracy, reliability, and low power consumption make it ideal for real-time clocks, data logging, and scheduling applications.

3.7 Software used

Blynk is a powerful IoT-based platform designed for developing mobile and web applications to control and monitor hardware projects remotely. It provides an intuitive and user-friendly interface that allows seamless interaction between hardware devices and smartphones through the internet. With its drag-and-drop widget system, Blynk makes it easy to create custom dashboards and integrate multiple features without requiring complex coding on the mobile side.

In the MediGuard 2.0 project, Blynk is used as the primary software tool for app development and medicine management. The application enables caregivers to schedule medication times, set reminders, and manage refilling operations directly from their smartphones. All essential functionalities—including time settings, notifications, and real-time updates—are accessible through the app, ensuring greater convenience and control. Caregivers can receive instant alerts when a medicine dose is due, missed, or delayed, thereby reducing supervision efforts and improving patient adherence to medication schedules.

The integration of Blynk also enhances scalability, as the system can be upgraded with additional features like cloud storage of medication history, remote access, and integration with voice assistants or telemedicine platforms. This flexibility makes Blynk an ideal choice for building secure, interactive, and efficient healthcare solutions like MediGuard 2.0.

Chapter 4

Conclusion

The MediGuard 2.0 project is a secure and automated healthcare solution designed to address missed or irregular medication intake among elderly, specially-abled, and chronically ill patients. It incorporates smart features such as fingerprint authentication for patients, dual authentication for caregivers, app-based monitoring, solenoid locks for security, and timely visual and audio reminders, ensuring accurate medication delivery and reducing human error. By combining electronics, automation, and healthcare, the system reduces caregiver workload while promoting patient independence and safety, and its scalable design allows use in both home and healthcare settings. The device also supports multiple users, tracks medicine stock levels, and maintains a log of medication history, providing both patients and caregivers with transparency and accountability. Looking ahead, MediGuard 2.0 can be enhanced with mobile app connectivity for real-time notifications, remote access, and medication tracking, as well as IoT and cloud integration to analyze patient data and support improved treatment decisions. Additional features like multilingual voice guidance, emergency alerts, telemedicine support, and predictive reminders can transform it into a comprehensive smart healthcare assistant capable of addressing diverse patient needs. Overall, MediGuard 2.0 demonstrates how technology-driven solutions can improve medication adherence, support caregivers, enhance patient safety, and ensure more reliable healthcare delivery, particularly for aging populations and remote care environments, while paving the way for future innovations in smart healthcare systems.

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